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## MID SEMESTER EXAMINATION Sep 2025

Name of the Course: B.Tech Aerospace Engineering

Semester: III

Name of the Paper: Aero Fluid Mechanics

Course Code: TAS-304

Time: 1-1/2Hour

Maximum Marks: 50

**Note:**(iii) Answer **all the questions** by choosing **any one of the sub questions**.

(iv) Each question carries 10 marks

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                            |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| <b>Q1</b> | (10 marks)                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                            |
| (a)       | Explain the following with their SI Unit<br>i. Continuity equation<br>ii. Kinematic Viscosity<br>iii. Weight density<br>iv. Velocity potential<br>v. Surface tension                                                                                                                                                                                                                                                                                                | <b>CO 1</b>                |
| <b>OR</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                            |
| (b)       | Define the Newton's law of Viscosity and the types of fluids? Hence Calculate the viscosity of the fluid between the plates where a plate 0.15mm distant from a fixed plate moved at 1.5 m/s and requires a force of 2.25 N/m <sup>2</sup> to maintain this speed?                                                                                                                                                                                                  |                            |
| <b>Q2</b> | (10 marks)                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                            |
| (a)       | Derive an expression for the pressure in terms of surface tension for:<br>i. Soap bubble<br>ii. Liquid jet                                                                                                                                                                                                                                                                                                                                                          | <b>CO 2</b>                |
| <b>OR</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                            |
| (b)       | Explain the phenomena of capillarity and hence the capillary rise and capillary depression? Derive an expression for the capillary depression in the capillarity?                                                                                                                                                                                                                                                                                                   |                            |
| <b>Q3</b> | (10 marks)                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                            |
| (a)       | Define the hydrostatic law? Derive an expression for pressure variation in fluid at rest using the hydrostatic law?                                                                                                                                                                                                                                                                                                                                                 | <b>CO 2</b>                |
| <b>OR</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                            |
| (b)       | What are the different pressure measurement devices? Discuss Piezometer and U-tube manometer in details with suitable diagram                                                                                                                                                                                                                                                                                                                                       |                            |
| <b>Q4</b> | (10 marks)                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                            |
| (a)       | A U-tube differential manometer containing mercury is connected on one side to pipe A containing CCl <sub>4</sub> (Sp. Gr. 1.5) under a pressure of 125kPa, and on the other side to pipe B containing oil (Sp. Gr. 0.85) under a pressure of 225kPa. The pipe A lies 2.5 m above pipe B and the mercury level in the limb communicating with pipe A lies 4 m below the pipe A. Determine the difference in the level of mercury in the two limbs of the manometer. | <b>CO 1</b><br><b>CO 2</b> |
| <b>OR</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                            |
| (b)       | A pipe (1) 450mm in diameter branches into two pipes (2 and 3) of diameter 300mm and 200mm respectively. If the average velocity in 450 mm diameter pie is 3m/s find?<br>i. Discharge through 450 mm diameter pipe                                                                                                                                                                                                                                                  | <b>CO 3</b>                |

|           |                                                                                                                                                        |             |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
|           | 300mm pipe is 2m/s,                                                                                                                                    |             |
| <b>Q5</b> | (10 marks)                                                                                                                                             | <b>CO 2</b> |
| (a)       | Discuss the different types of fluid flow in kinematic flow of fluid?<br>Explain those by the help of mathematical expression?                         |             |
|           | <b>OR</b>                                                                                                                                              |             |
| (b)       | Explain the following and discuss them with proper mathematical expression:<br>i. Velocity Potential<br>ii. Equipotential line<br>iii. Stream function |             |